RAJALAKSHMI ENGINEERING COLLEGE

Rishi K 251001211-2025-IT

R2

Started on	Sunday, 16 August 2026, 10:36 PM
State	Finished
Completed on	Sunday, 16 August 2026, 10:37 PM
Time taken	28 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findFloor(int arr[], int low, int high, int x) {
    // Base case: no valid range
    if (low > high) {
        return -1;
    }

    // If x is smaller than the smallest element, no floor exists
    if (x < arr[low]) {
        return -1;
    }

    // If x is greater than or equal to the largest element, floor is the last element
    if (x >= arr[high]) {
        return arr[high];
    }

    int mid = low + (high - low) / 2;

    if (arr[mid] == x) {
        return arr[mid];
    }

    // If mid is the second-last check: arr[mid] < x < arr[mid+1], floor is arr[mid]
    if (mid < high && arr[mid] < x && arr[mid + 1] > x) {
        return arr[mid];
    }

    if (arr[mid] > x) {
        // floor lies in left half
        return findFloor(arr, low, mid - 1, x);
    } else {
        // arr[mid] < x, floor lies in right half
        return findFloor(arr, mid + 1, high, x);
    }
}

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    scanf("%d", &x);

    int result = findFloor(arr, 0, n - 1, x);
    printf("%d\n", result);
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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